

KEVIN HUANG

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SUMMARY

Electrical Engineering PhD student at the University of Southern California studying speech production with the Signal Analysis and Interpretation Laboratory (SAIL). Over a decade of coding experience with a strong background in Machine Learning. Research focuses on articulatory phonetics from real-time MRI, with broader interests spanning audio processing and music information retrieval. Multiple publications at top-tier venues, with two Best Student Paper Awards.

EXPERIENCE

Research Assistant

University of Southern California — SAIL

📅 August 2023 – Present

- Research in the Signal Analysis and Interpretation Laboratory and the interdisciplinary SPAN group, studying speech production by collecting and analyzing real-time MRI data.
- Involved in projects spanning speech foundation model benchmarking and multi-modal speech and neural signal processing

Computer Science TA / Tutor

University of California San Diego

📅 April 2018 – March 2021

- Teaching Assistant for Computer Graphics (Fall 2020) and Computer Animation (Winter 2021).
- Tutor for 7 quarters and 2 summer sessions across Discrete Math, Digital Systems, and Computer Graphics.

Software Engineering Intern

Webroot Cybersecurity

📅 June 2018 – September 2018 📍 San Diego, CA

- Revamped a DNS agent distribution portal from scratch.
- Built an Angular front-end with a Node.js back-end connected to a MySQL database.

SELECTED PUBLICATIONS

Huang, K., Goldberg, J., Goldstein, L., & Narayanan, S. "Analysis of articulatory setting for L1 and L2 English speakers using MRI data." *Interspeech* 2024. **Best Student Paper Award**

Huang, K., Foley, S., Lee, J., Lee, Y., Byrd, D., & Narayanan, S. "On the Relationship between Accent Strength and Articulatory Features." *Interspeech* 2025. **Best Student Paper Award**

Lee, J., Razmara, P., Huang, K., et al. "An Approach to Simultaneous Acquisition of Real-Time MRI Video, EEG, and Surface EMG for Articulatory, Brain, and Muscle Activity During Speech Production." *Interspeech* 2026.

EDUCATION

PhD Electrical Engineering

University of Southern California

📅 August 2022 – Present

- Signal Processing track, Linguistics minor

MS Computer Science

University of California San Diego

📅 Sept 2020 – June 2021

- Emphasis on Machine Learning

BS Computer Science

University of California San Diego

📅 Sept 2016 – June 2020

- Minors: Mathematics; Music Computing

SKILLS

- **Languages:** Python, C++, Javascript, C, Java
- **Libraries:** PyTorch, TensorFlow, OpenGL, React
- **Software:** Git, Pure Data, Ableton, Blender

SELECT PROJECTS

RASPPPy

📅 Fall 2023 – Present

- Visual programming language for prototyping real-time audio processing apps.
- Written in Python and JavaScript.

Klatt Voice Synthesizer

📅 Winter 2020

- MIDI-controllable source-filter voice synthesizer with real-time queueable phonemes.
- Built for STM32F4 microprocessor in C; prototyped with Pure Data.